Practical 1: SSL/TLS Configuration Testing

1. Objective

Evaluate and test the configuration of SSL/TLS for web applications to ensure secure data transmission.

2. Tools Required

- Qualys SSL Labs Scanner

3. Theory

SSL/TLS protocols are responsible for encrypting data transmission between clients and servers. Misconfigurations such as outdated protocols (SSLv2, SSLv3, TLS 1.0) or weak cipher suites can lead to vulnerabilities like Man-in-the-Middle (MITM) attacks. By using tools such as Qualys SSL Labs Scanner, we can evaluate the security strength of SSL/TLS configurations and ensure they follow best practices.

4. Steps to Perform SSL/TLS Configuration Testing

1. Open a browser and go to https://www.ssllabs.com/ssltest/.

2. Enter the domain name of the target web application (e.g., example.com).

3. Click on the 'Submit' button to start the SSL/TLS configuration scan.

4. Wait for the scanner to complete the analysis (this may take a few minutes).

5. Review the detailed report generated by Qualys SSL Labs.

6. Identify weak ciphers, outdated SSL/TLS protocols, or incorrect certificate configurations.

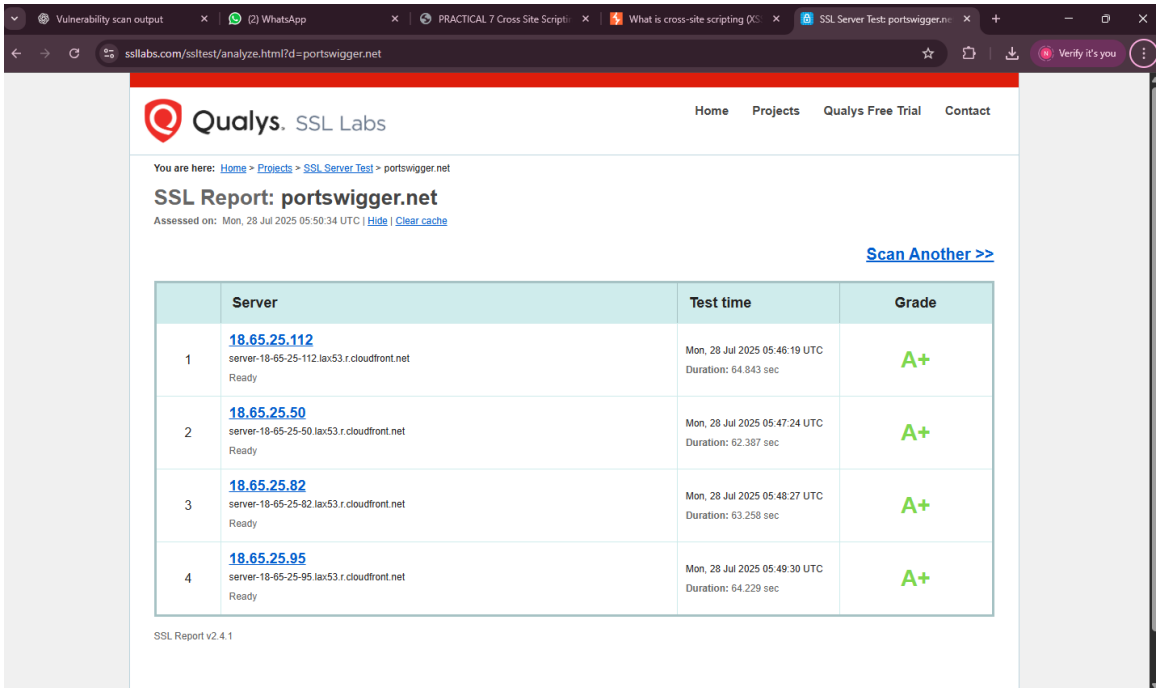
7. Note the overall grade (A+, A, B, etc.) assigned to the server.

8. Based on the findings, suggest improvements such as disabling outdated protocols and using strong cipher suites (e.g., TLS 1.2 or TLS 1.3 with AES-256).

5. Expected Outcome

- Identification of weak ciphers (e.g., RC4, 3DES).
- Detection of outdated protocols (SSLv2, SSLv3, TLS 1.0).

• Suggestions to upgrade to stronger ciphers and TLS 1.2/1.3.
• Improved SSL/TLS configuration with an A or A+ grade.



6. Conclusion

By performing SSL/TLS configuration testing using Qualys SSL Labs Scanner, we can ensure that the web application uses modern encryption standards, thereby protecting data from interception and attacks.